

MLProvLens: Exploring End-to-End provenance in ML Pipelines with a W3C PROV-Aligned Framework

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Abstract. Once specified and enacted, end-to-end machine learning (ML) pipelines, together with contextual information about how artifacts are consumed and produced during execution, constitute valuable assets that can be shared or published for reuse. Indeed, ML pipelines involve multiple repetitive data transformation steps before, during, and after model training. Evaluating alternative configurations at each stage requires continuous monitoring of the workflow, particularly to support informed model selection for deployment. Although existing monitoring solutions record metrics and configurations, these logs are typically expressed using ad hoc data models, and lack support for tracing complete derivation paths from training data to deployed models. To overcome such limitation, we introduced in a previous work a unified, W3C PROV-compatible meta-model for capturing end-to-end provenance of ML pipelines. In this demonstration, we present an interactive system that operationalizes this meta-model through a lightweight provenance exploration interface. The graphical user interface allows user to load a provenance graph, search for nodes of interest, render provenance sub-graphs, inspect node metadata and relationships, and execute predefined end-to-end provenance queries over the full pipeline.

Keywords: Provenance · ML pipelines · W3C PROV · Interactive GUI.

1 Introduction

The development of machine-learning (ML) models is inherently exploratory and iterative, spanning data ingestion, preprocessing, feature engineering, model design, training, and optimisation. Because predictive performance relies heavily on upstream data transformations and training configurations, understanding an ML pipeline requires reasoning over its end-to-end flow rather than isolated steps.

Provenance provides a formal record of the origin, context, and derivation history of data products and artifacts. In complex ML pipelines, effective provenance tracking must cover the entire lifecycle while handling heterogeneous artifact granularities. Existing approaches, however, address only parts of this problem. Fine-grained frameworks capture preprocessing provenance at the level of records

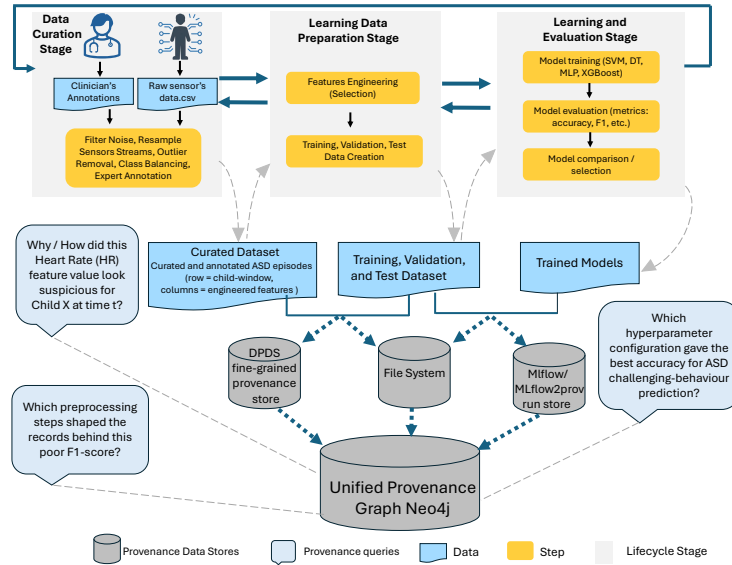


Fig. 1: End-to-end ML pipeline lifecycle and provenance integration.

and individual values, but remain limited to the preprocessing phase [2]. In contrast, MLflow2PROV tracks runs, parameters, models, and related experiment artifacts, yet lacks a unified view of upstream data transformations [3].

Our system bridges the gap between these disparate tracking tools by integrating fine-grained data-preparation provenance with coarse-grained experiment tracking into a unified graph. Through an interactive GUI, users can explore provenance graphs, inspect artifact metadata, and execute end-to-end queries that trace model evaluation metrics back to their originating data transformations.

2 System Architecture

Our demonstration system operationalizes the provenance meta model introduced in our previous work [1], for capturing end-to-end provenance in ML pipelines. The model distinguishes between prospective lineage, which describes the intended pipeline structure at design time, and retrospective lineage, which captures the actual execution history. This separation enables unified representation that connects the planned workflow with its execution, providing complete traceability of how data, models, and results are used and generated through the ML lifecycle. A detailed description of our meta-model is presented in [1].

ML pipelines are designed to be centered around three stages: Data Curation, Learning Data Preparation, and Learning. As depicted in Fig. 1, the Data Curation stage transforms raw inputs through operations such as filtering, resampling, outlier removal, and annotation. The learning data preparation stage refines the curated data through feature engineering and splitting. Finally, the Learning Stage iteratively perform model comparison and selection, producing

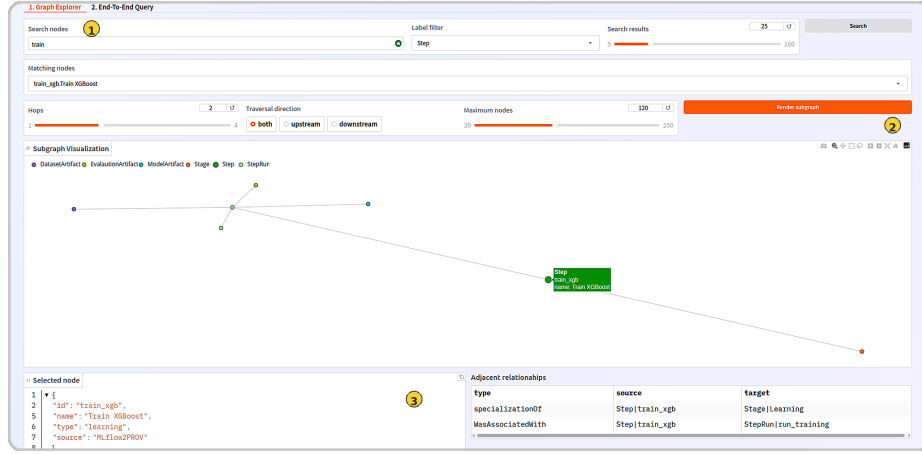


Fig. 2: Screenshot of part of the demonstrator. (1) The user searches for a provenance node and selects a matching result. (2) The user configures traversal parameters and renders the corresponding subgraph. (3) The system displays detailed metadata and adjacent relationships for the selected node.

optimized trained models and performance metrics. The figure also highlights how provenance queries can trace model outputs back to their originating data transformations.

To capture provenance across these stages, our system integrates two complementary provenance solutions: DPDS for fine-grained preprocessing provenance [2], and MLflow2PROV for experiment and model lifecycle provenance [3]. These are unified into a single provenance graph.

To support interactive exploration, the system provides a modular GUI architecture. The unified graph is exported into `nodes.csv` and `rels.csv`, and indexed in a local SQLite database for efficient querying. The data-access layer executes provenance queries over this index database, while the control layer, implemented using Gradio, manages user interaction. Finally, the visualization layer, based in Plotly and NetworkX, renders the interactive provenance graph and associated charts.

3 Demonstration Scenarios

Initially, the user loads the provenance graph by providing the `nodes.csv` and `rels.csv` files generated by the provenance meta-model implementation. Once loaded, the system initializes (or reuses) the SQLite index and displays a statistical summary of the graph structure, including the counts of nodes and relationships. The demonstrator then proceeds with the following two scenarios.

Graph Exploration: In this scenario, the user (e.g., a clinician) explores a provenance fragment centered on a pipeline element of interest. As shown in Fig. 2 (1), the user enters a keyword in the search field, optionally filters by node type, and retrieves matching nodes. After selecting a node, the user configures

traversal depth and direction, and renders the corresponding subgraph (Fig. 2 (2)). Upstream traversal reveals lineage, downstream traversal shows impact, and bidirectional traversal exposes the local neighborhood. To ensure interactive rendering, the user may limit the number of displayed nodes. Fig. 2 illustrates an example centered on the training step `train_xgb`. Additional metadata and adjacent relationships for the selected node are displayed in the inspection panel (Fig. 2 (3))

End-to-End Query Execution: The GUI also provides a second tab, End-To-End Query, where the user selects a query from a predefined list of 15 provenance queries introduced in our previous work [1]. These queries cover data curation, data-to-model integration, and learning/configuration analysis. After selecting a query and providing the required input parameters, the system executes the corresponding provenance traversal over the unified graph. Results are presented both as a textual explanation and as a highlighted subgraph. For example, a user may trace a poor F1-score back to the preprocessing steps and records that influenced the training data.

For demonstration purposes, the GUI operates on a reduced provenance graph to ensure readability and responsiveness during live interaction. The underlying meta-model and implementation, however, have been evaluated on real datasets as reported in our previous work ¹.

4 Conclusions and Future Work

This demonstration complements our previous provenance meta-model by providing an interactive system for exploring and querying end-to-end provenance in ML pipelines. While the meta-model integrates fine-grained preprocessing provenance with experiment-level model provenance, the system enables users to move seamlessly from raw and curated data artifacts to model runs, trained models, and evaluation outcomes. This makes the demo particularly relevant for provenance-driven debugging, model selection, and inspection of complex ML workflows. ongoing work focuses on applying and assessing the demonstrator on a broader range of pipelines, as well as extending the query capabilities.

References

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¹ The source code is available at https://github.com/aq1992dauphine/End_To_End_Provenance_Meta_Model. A video walkthrough illustrating the complete demonstration scenarios, along with a technical report are available at: https://drive.google.com/drive/folders/1rIVZiES0V18ay0kfXXWm7ytVBcdC_KZj?usp=sharing.